Skylar AI - Kubernetes Cluster Sizing Guide

Overview

The Skylar platform provides four pre-configured scaling profiles based on **Data Points Per Minute (DPM)** - the primary metric for determining cluster sizing requirements. DPM is measured by querying the ClickHouse OTEL metrics table for new rows created.

DPM-Based Sizing Guidelines

Performance Targets

- **Maximum Acceptable Delay**: 20 minutes for artifact generation (reporting data, anomalies, predictive alerts)
- **Monitoring**: Skylar Monitoring Service provides `clickhouse_datapoints_per_minute` and `clickhouse_cluster_datapoints_per_minute` metrics
- **Measurement Period**: Rolling 24-hour count (new clusters take 24 hours for accurate counting)

Complete Scaling Profile Comparison

Small Deployment

DPM Range: 0 - 30,000 DPM
Target Use Case: Development, QA, and penetration testing environments

Core Infrastructure

Component	CPU Request	Memory Request	Storage	Notes
ClickHouse	4 cores	16Gi	350Gi	Single replica
PostgreSQL HA	250m	7Gi	128Gi	1 PgPool replica, 410 max connections

ZooKeeper	100m	1Gi	Default	Minimal configuration

Key Skylar Services

Service	CPU Request	Memory Request	Notes
Metadata Snapshot	250m	2Gi	Core analytics service
Policy Syncer UI	50m	1Gi	AI/ML policy management
Policy Syncer Alert	50m	1Gi	Alert processing
Event Viz API	50m	1Gi	Event visualization
Platform Services	150m	500Mi	Core platform
Skylar Auth (Hera)	50m	512Mi	Authentication
Skylar Auth (UI)	100m	500Mi	Auth UI
Reporting Service	50m	500Mi	Report generation
Monitoring Service	20m	64Mi	System monitoring

AI/ML Services

Service	CPU Request	Memory Request	Notes
AIML Anomaly Detection	200m total	2.5Gi total	4 components
Vitals CPU Prediction	50m	1Gi	CPU prediction API

SLS API AIML Policy	50m	200Mi	Policy API
SLS API Anomaly Alerts	50m	300Mi	Alert API

Estimated Total: ~8-10 cores, ~30-35Gi RAM, ~480Gi storage

Medium Deployment

DPM Range: 30,000 - 215,000 DPM

Target Use Case: Small production environments or staging

Core Infrastructure

Component	CPU Request	Memory Request	Storage	Notes
ClickHouse	16 cores	128Gi	512Gi	Mid-scale analytics
PostgreSQL HA	250m	13Gi	128Gi	3 PgPool replicas, 1210 max connections
ZooKeeper	400m	2Gi	Default	Enhanced configuration

Key Skylar Services

Service	CPU Request	Memory Request	Notes
Metadata Snapshot	4 cores	4Gi	Enhanced processing
Policy Syncer UI	500m	6Gi	Increased AI workload
Policy Syncer Alert	500m	6Gi	Enhanced alert processing

Event Viz API	50m	2Gi	Higher memory for startup
Platform Services	150m	500Mi	Core platform
Skylar Auth (Hera)	50m	512Mi	Authentication
Skylar Auth (UI)	150m	500Mi	Enhanced auth UI
Reporting Service	50m	500Mi	Report generation
Monitoring Service	50m	64Mi	Enhanced monitoring

AI/ML Services

Service	CPU Request	Memory Request	Notes
AIML Anomaly Detection	200m total	3Gi total	Enhanced OLAP component
Vitals CPU Prediction	50m	1Gi	CPU prediction API
SLS API AIML Policy	50m	200Mi	Policy API
SLS API Anomaly Alerts	50m	300Mi	Alert API

Estimated Total: ~25-30 cores, ~170-180Gi RAM, ~640Gi storage

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Large Deployment

DPM Range: 215,000 - 300,000 DPM

Target Use Case: Production environments with moderate to high load

Core Infrastructure

Component	CPU Request	Memory Request	Storage	Notes
ClickHouse	32 cores	256Gi	512Gi	High-performance analytics
PostgreSQL HA	1 core	24Gi	128Gi	3 PgPool replicas, 1210 max connections
ZooKeeper	1 core	4Gi	32Gi	Production configuration

Key Skylar Services

Service	CPU Request	Memory Request	Notes
Metadata Snapshot	4 cores	4Gi	Production processing
Policy Syncer UI	400m	16Gi	High-scale AI processing
Policy Syncer Alert	400m	16Gi	Production alert processing
Event Viz API	50m	2Gi	Production visualization
Platform Services	150m	500Mi	Core platform
Skylar Auth (Hera)	50m	512Mi	Authentication
Skylar Auth (UI)	150m	500Mi	Auth UI
Reporting Service	50m	500Mi	Report generation

Monitoring Service	100m	256Mi	Production monitoring

AI/ML Services

Service	CPU Request	Memory Request	Notes
AIML Anomaly Detection	200m total	3Gi total	Production AI processing
Vitals CPU Prediction	50m	1Gi	CPU prediction API
SLS API AIML Policy	50m	200Mi	Policy API
SLS API Anomaly Alerts	50m	300Mi	Alert API

Estimated Total: ~45-50 cores, ~330-340Gi RAM, ~690Gi storage

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XLarge Deployment

DPM Range: 300,000 - 900,000 DPM
Target Use Case: High-scale production environments with heavy workloads
Max Single Instance: 977,000 DPM (observed)

Core Infrastructure

Component	CPU Request	Memory Request	Storage	Notes
ClickHouse	64 cores	512Gi	4Ti	Massive data processing
PostgreSQL HA	1 core	13Gi	512Gi	3 PgPool replicas, 1210 max connections
ZooKeeper	3 cores	2Gi	32Gi	High availability

Key Skylar Services

Service	CPU Request	Memory Request	Notes
Metadata Snapshot	2 cores	4Gi	Optimized for scale
Policy Syncer UI	2 cores	24Gi	Maximum AI processing
Policy Syncer Alert	1 core	16Gi	High-scale alert processing
Event Viz API	250m	2Gi	Enterprise visualization
Platform Services	150m	500Mi	Core platform
Skylar Auth (Hera)	50m	512Mi	Authentication
Skylar Auth (UI)	150m	500Mi	Auth UI
Reporting Service	50m	500Mi	Report generation
Monitoring Service	100m	256Mi	Enterprise monitoring

AI/ML Services

Service	CPU Request	Memory Request	Notes
AIML Anomaly Detection	200m total	3Gi total	Enterprise AI processing
Vitals CPU Prediction	50m	1Gi	CPU prediction API
SLS API AIML Policy	50m	200Mi	Policy API

SLS API Anomaly Alerts	50m	300Mi	Alert API

Estimated Total: ~80-85 cores, ~580-590Gi RAM, ~4.5Ti storage

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Detailed Resource Scaling Patterns

ClickHouse Scaling (Primary Data Store)

Size	CPU	Memory	Storage	Scale Factor
Small	4 cores	16Gi	350Gi	1x
Medium	16 cores	128Gi	512Gi	4x CPU, 8x RAM
Large	32 cores	256Gi	512Gi	8x CPU, 16x RAM
XLarge	64 cores	512Gi	4Ti	16x CPU, 32x RAM, 11x storage

Policy Syncer Services (Memory-Intensive AI/ML)

Size	CPU (UI/Alert)	Memory (UI/Alert)	Notes
Small	50m / 50m	1Gi / 1Gi	Minimal AI processing
Medium	500m / 500m	6Gi / 6Gi	Moderate AI workloads
Large	400m / 400m	16Gi / 16Gi	Production AI processing
XLarge	2000m / 1000m	24Gi / 16Gi	High-scale AI/ML

PostgreSQL HA Configuration

Size	PgPool Replicas	Max Connections	Memory	Storage
Small	1	410	7Gi	128Gi
Medium	3	1210	13Gi	128Gi
Large	3	1210	24Gi	128Gi
XLarge	3	1210	13Gi	512Gi

Metadata Snapshot Service (Core Analytics)

Size	CPU	Memory	Notes
Small	250m	2Gi	Basic processing
Medium	4 cores	4Gi	Enhanced processing
Large	4 cores	4Gi	Production processing
XLarge	2 cores	4Gi	Optimized for scale

Node Requirements & Architecture

Recommended Node Pool Strategy

Database Node Pool

- **Node Type**: High-memory instances (e.g., r5.4xlarge or larger)
- **Node Selector**: `type: database`
- **Tolerations**: `workload=database`
- **Purpose**: ClickHouse, PostgreSQL workloads

Application Node Pool

- **Node Type**: Balanced compute instances (e.g., m5.2xlarge)
- **Purpose**: Microservices, AI/ML processing, web services

Storage Requirements

- **Storage Classes**: Use your environment's preferred storage class (e.g., `gp3`, `fast-ssd`, `premium-storage`)
- **Performance**: Ensure high-IOPS storage for ClickHouse workloads
- ⚠️ **Critical**: **DO NOT use NFS-backed storage classes** - ClickHouse and PostgreSQL require direct-attached or block storage for optimal performance and data integrity
- **Backup Strategy**: Ensure adequate backup storage for ClickHouse (especially XLarge: 4Ti)

Minimum Cluster Recommendations

Deployment Size	DPM Range	Min Nodes	Recommended Instance Types	Total vCPUs	Total RAM
Small	0-30k	2-3 nodes	1x m5.xlarge + 1x r5.xlarge	8-12	32-48Gi
Medium	30k-215k	3-4 nodes	1x m5.2xlarge + 2x r5.4xlarge	24-32	128-192Gi
Large	215k-300k	4-6 nodes	2x m5.4xlarge + 2x r5.8xlarge	48-64	256-384Gi
XLarge	300k-900k	6-8 nodes	2x m5.8xlarge + 3x r5.12xlarge	80-96	512-768Gi

Critical Considerations

1. **DPM is Primary Sizing Metric**: Use DPM thresholds to determine appropriate cluster size
2. **AIML Poller is Critical Path**: This service determines overall system lag and performance

3. **Memory-Intensive AI/ML**: Policy Syncer services require significant RAM scaling (1Gi → 24Gi)
4. **ClickHouse Dominates Resources**: In larger deployments, ClickHouse consumes 70-80% of total cluster resources
5. **Storage Growth**: XLarge deployments require 11x more storage than Small
6. **Network**: Ensure adequate network bandwidth for data-intensive workloads
7. **Storage Performance**: Use high-IOPS storage classes for optimal ClickHouse performance
8. ⚠️ **Storage Type**: **Avoid NFS-backed storage** - Use block storage (EBS, Azure Disk, GCE Persistent Disk) for database workloads
9. **Monitoring**: Use Skylar Grafana Dashboard to track tenant and cluster-level DPMs
10. **Resource Monitoring**: Monitor services using >80% of provisioned resources to prevent OOM events

Service Categories Summary

Core Infrastructure (75-85% of resources)

- ClickHouse (primary data store)
- PostgreSQL HA (metadata and configuration)
- ZooKeeper (coordination)

Analytics & AI/ML (10-20% of resources)

- Metadata Snapshot Service
- Policy Syncer Services
- AIML Anomaly Detection
- Vitals CPU Prediction

Platform Services (5-10% of resources)

- Authentication services
- Platform management
- Reporting
- Monitoring

This sizing guide provides the foundation for capacity planning and infrastructure provisioning for Skylar platform deployments based on actual DPM performance data and resource requirements.